

BIKE RENTAL EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis Using Python

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Technologies Used

Python • Pandas • Matplotlib • VS Code

BIKE RENTAL EXPLORATORY DATA ANALYSIS

1. Introduction & Objective

Introduction

Bike sharing is a transportation system in which people can rent bicycles for short trips. The number of bike rentals can change depending on factors such as weather, season, month, day of the week, working day, and time of day.

In this project, the **Bike Sharing Dataset** is analyzed to understand patterns and relationships between bike rentals and different weather, seasonal, and time-related factors.

The dataset contains daily and hourly bike rental information collected from the **Capital Bikeshare system during 2011 and 2012**.

Exploratory Data Analysis (EDA) techniques are used to understand the dataset, check its quality, calculate statistical measures, identify patterns, study relationships between variables, and present the results using graphs and charts.

Objectives

The main objectives of this project are:

1. To understand and describe the bike rental dataset.
 2. To check the quality of the data, including missing values, duplicate records, invalid values, and data consistency.
 3. To calculate descriptive statistics such as mean, median, minimum, maximum, and standard deviation.
 4. To study individual variables using univariate analysis.
 5. To study relationships between variables using bivariate analysis.
 6. To compare bike rentals across seasons, months, years, weather conditions, and working days.
 7. To analyze relationships between weather variables and bike rentals using correlation analysis.
 8. To create suitable visualizations such as histograms, bar charts, line charts, scatter plots, boxplots, and correlation heatmaps.
 9. To identify important patterns and findings from the data.
 10. To summarize the analysis and present the conclusions clearly.
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2. Dataset Description

2.1 Dataset Source

The dataset used in this project is the **Bike Sharing Dataset** obtained from the **UCI Machine Learning Repository**.

The dataset contains bike rental information collected from the Capital Bikeshare system during **2011 and 2012**. It includes information about bike rentals, weather conditions, seasons, months, days, and user types.

The dataset contains two main files:

- **day.csv** – daily bike rental information.
- **hour.csv** – hourly bike rental information.

The local files used in this project were analyzed using Python and Pandas.

The local day.csv file contains **731 records and 16 variables**, while the local hour.csv file contains **17,379 records and 17 variables**.

The data quality of these files was checked during the analysis, including missing values, duplicate records, data types, valid categorical values, and consistency of total rental counts.

2.2 Files Used

day.csv

The day.csv file contains information about bike rentals recorded on a daily basis.

It is useful for studying:

- Daily rental demand
- Seasonal patterns
- Monthly patterns
- Weather conditions
- Working days and holidays
- Casual and registered users

hour.csv

The hour.csv file contains information about bike rentals recorded on an hourly basis.

It contains an additional hr variable, which represents the hour of the day.

It can be used to study:

- Hourly rental patterns
- Peak rental hours
- Time-based rental demand
- Hourly weather and rental relationships

2.3 Important Variables

Variable	Meaning
Instant	Record/index number
Dteday	Date
Season	Season of the year
Yr	Year code: 0 = 2011, 1 = 2012
Mnth	Month
Hr	Hour of the day; available in hour.csv
Holiday	Indicates whether the day is a holiday
Weekday	Day of the week
workingday	Indicates whether it is a working day
weathersit	Weather condition
Temp	Normalized temperature
Atemp	Normalized feeling temperature
Hum	Normalized humidity
windspeed	Normalized wind speed
Casual	Number of casual users
Registered	Number of registered users
Cnt	Total number of rented bikes

The cnt variable represents the total number of rented bikes and is calculated from casual and registered users.

$$\text{cnt} = \text{casual} + \text{registered}$$

2.4 Dataset Structure

The two datasets were used for different purposes.

day.csv → Daily Analysis

Examples:

- Rentals by season
- Rentals by month
- Rentals by weather
- Daily rental demand
- Weather vs rental relationship

hour.csv → Hourly Data Validation

The hourly dataset was used mainly for dataset inspection and consistency checking in this project. It also contains the hr variable, which can be used for detailed hourly analysis.

The combination of daily and hourly data provides information at both daily and hourly levels.

3. Data Types and Variables

The Bike Sharing dataset contains numerical, categorical, binary, and date variables.

Variable	Description	Data Type
Instant	Record/index number	Integer
Dteday	Date of the record	Date
Season	Season of the year	Categorical
Yr	Year	Categorical
Mnth	Month	Categorical
Hr	Hour of the day	Numerical / Ordered
Holiday	Holiday indicator	Binary
Weekday	Day of the week	Categorical

Variable	Description	Data Type
workingday	Working day indicator	Binary
weathersit	Weather condition	Categorical
Temp	Normalized temperature	Numerical
Atemp	Normalized feeling temperature	Numerical
Hum	Normalized humidity	Numerical
windspeed	Normalized wind speed	Numerical
Casual	Number of casual users	Numerical
Registered	Number of registered users	Numerical
Cnt	Total number of rented bikes	Numerical

Types of Variables

3.1 Numerical Variables

Examples:

- temp
- atemp
- hum
- windspeed
- casual
- registered
- cnt

These variables contain measurements or numerical counts.

3.2 Categorical Variables

Examples:

- season
- yr
- mnth
- weekday

- weathersit

Although these variables are represented using numbers, the numbers represent categories or codes.

3.3 Binary Variables

Binary variables have two possible values.

Examples:

- holiday: 0 = No, 1 = Yes
- workingday: 0 = No, 1 = Yes

3.4 Date Variable

dteday represents the date on which the bike rentals were recorded.

3.5 Target Variable

cnt represents the total number of rented bikes and is the main variable used to study bike rental demand.

4. Tools Used for Analysis

The following tools and libraries were used in this project.

Tool / Library Purpose

Python	Main programming language
Pandas	Data loading, cleaning, grouping, aggregation, and analysis
Matplotlib	Creating graphs and visualizations
VS Code	Writing and executing Python code
GitHub	Storing and sharing the project code

Python

Python was used as the main programming language for performing the analysis.

Pandas

Pandas was used to:

- Read CSV files
- Inspect datasets

- Check missing values
- Check duplicate records
- Convert dates
- Calculate statistics
- Perform grouping and aggregation
- Calculate correlations

Matplotlib

Matplotlib was used to create:

- Histograms
- Scatter plots
- Bar charts
- Line charts
- Boxplots
- Correlation heatmaps

VS Code

VS Code was used as the development environment for writing and executing the Python code.

GitHub

GitHub can be used to store and share the final project source code.

5. Data Quality Assessment

Data quality checks were performed before the main analysis.

The following checks were performed:

1. Missing values
2. Duplicate rows
3. Data types
4. Invalid categorical values
5. Data consistency

6. Potential outliers

5.1 Missing Value Check

Missing values were checked using the Pandas `isnull().sum()` function.

The analysis showed that both the `day.csv` and `hour.csv` datasets contain **0 missing values** in all columns.

Therefore, no missing-value treatment or imputation was required.

Conclusion: No missing values were found in the datasets.

5.2 Duplicate Row Check

Duplicate records were checked using the Pandas `duplicated()` function.

The analysis found **0 duplicate rows** in the datasets.

Therefore, no duplicate records needed to be removed.

Conclusion: No duplicate records were found.

5.3 Data Type Check

The data types of the variables were examined using the Pandas `dtypes` function.

Most variables were stored as integer or floating-point values.

The `dteday` variable was initially stored as a string. Since it represents dates, it was converted to datetime format using:

```
pd.to_datetime()
```

This makes the date variable more suitable for time-based analysis.

5.4 Invalid Value Check

The categorical variables were checked to identify values outside their expected ranges.

The observed values were:

- season: 1 to 4
- yr: 0 and 1
- mnth: 1 to 12
- hr: 0 to 23
- weathersit: 1 to 4

All observed values were within their expected ranges.

Conclusion: No invalid categorical values were found in the checked variables.

5.5 Data Consistency Check

The relationship between casual, registered, and cnt was checked.

The expected relationship is:

casual + registered = cnt

The analysis found **0 incorrect rows among the 17,379 hourly records**.

Therefore, the total rental values were consistent with the sum of casual and registered users.

Conclusion: The cnt values were consistent with the component user counts.

5.6 Outlier Check

Boxplots were used to identify potential outliers in selected numerical variables.

Some observations appeared outside the main distribution, particularly for the registered and cnt variables.

These observations were not automatically treated as errors because unusually high or low rental counts may represent genuine periods of high or low bike rental demand.

Therefore, the observations were retained for further analysis.

6. Descriptive Statistics

Descriptive statistics were calculated using the daily bike rental dataset (day.csv).

The Pandas describe() function was used to summarize the numerical variables.

The daily dataset contains **731 observations**.

Important Results

Variable	Mean	Median	Minimum	Maximum
Casual users	848.18	713	2	3,410
Registered users	3,656.17	3,662	20	6,946
Total rentals (cnt)	4,504.35	4,548	22	8,714

Interpretation

The average number of total bike rentals per day was approximately **4,504**.

The average number of registered-user rentals was approximately **3,656**, while the average number of casual-user rentals was approximately **848**.

The maximum daily rental count was **8,714**, while the minimum was **22**.

The standard deviation of cnt was approximately **1,937.21**, indicating considerable variation in daily rental demand.

The weather variables such as temp, atemp, hum, and windspeed are represented using normalized values in the original dataset. Therefore, these values should not be directly interpreted as Celsius, percentage, or other physical units.

7. Univariate Analysis

Univariate analysis means analyzing one variable at a time.

Histograms were used to study the distribution of:

- Total bike rentals (cnt)
- Casual users (casual)
- Registered users (registered)
- Temperature (temp)
- Humidity (hum)
- Windspeed (windspeed)

7.1 Total Bike Rentals

The cnt variable represents the total number of bikes rented per day.

Measure	Value
Mean	4504.35
Median	4548
Minimum	22
Maximum	8714

Standard Deviation 1937.21

The average daily bike rental count was approximately **4,504 bikes**.

The minimum number of rentals was **22**, while the maximum was **8,714**.

The standard deviation of **1,937.21** shows considerable variation in daily rental demand.

7.2 Casual Users

The casual variable represents rentals made by casual users.

Measure	Value
---------	-------

Mean	848.18
------	--------

Median	713
--------	-----

Minimum	2
---------	---

Maximum	3410
---------	------

The average number of casual-user rentals was approximately **848 per day**.

The number of casual users varied from **2 to 3,410**.

7.3 Registered Users

The registered variable represents rentals made by registered users.

Measure	Value
---------	-------

Mean	3656.17
------	---------

Median	3662
--------	------

Minimum	20
---------	----

Maximum	6946
---------	------

The average number of registered-user rentals was approximately **3,656 per day**.

The number varied from **20 to 6,946**.

7.4 Weather Variables

Temperature, humidity, and windspeed were also analyzed using histograms.

These graphs help understand the distribution of weather conditions recorded in the dataset.

7.5 Key Observation

The univariate analysis shows that bike rental demand varies considerably from day to day.

Registered users contributed a larger average number of rentals than casual users.

The weather variables also showed variation across the recorded days.

8. Bivariate Analysis

Bivariate analysis was used to study the relationship between two variables.

8.1 Temperature vs Total Bike Rentals

A scatter plot was used to compare temperature (temp) with total bike rentals (cnt).

The correlation between temperature and total bike rentals was:

0.6275

This indicates a **moderate positive relationship**.

In general, higher temperatures were associated with higher numbers of bike rentals in the dataset.

However, correlation does not prove that temperature alone causes changes in bike rental demand.

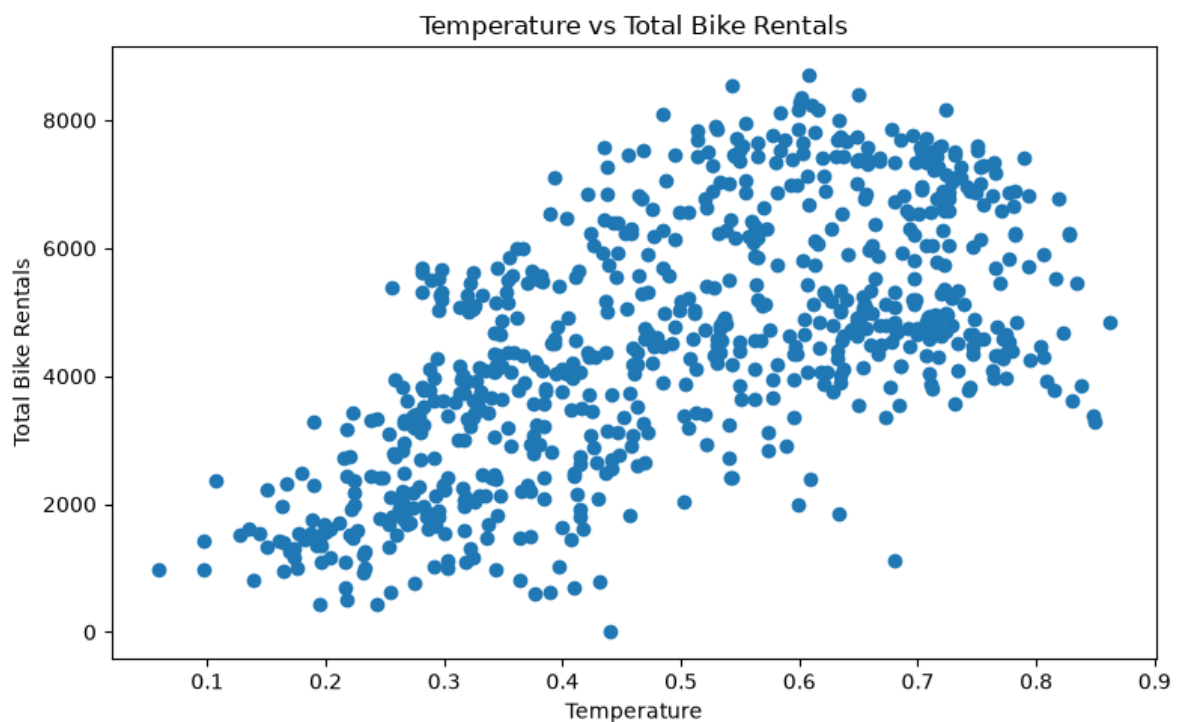


Figure 8.1: Temperature vs Total Bike Rentals

8.2 Humidity vs Total Bike Rentals

A scatter plot was used to compare humidity (hum) with total bike rentals (cnt).

The correlation was:

-0.1007

This indicates a **weak negative relationship**.

Higher humidity was slightly associated with lower bike rental counts in the dataset.

However, the relationship is weak and does not prove causation.

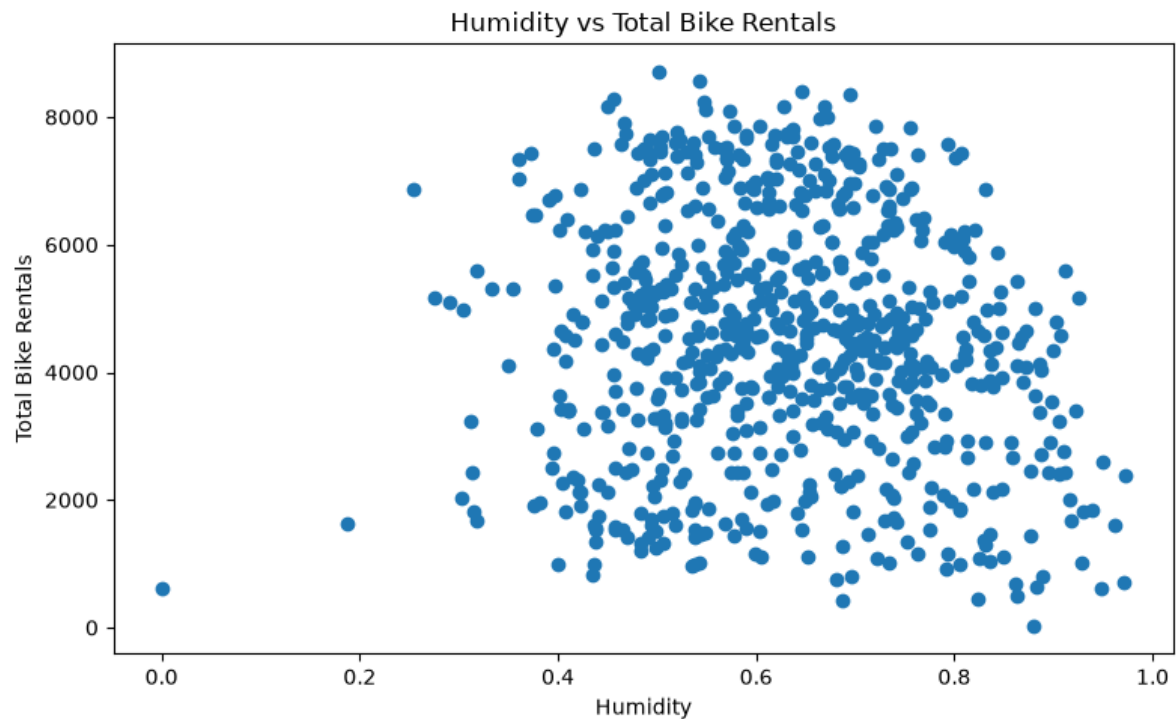


Figure 8.2: Humidity vs Total Bike Rentals

8.3 Windspeed vs Total Bike Rentals

A scatter plot was used to compare windspeed with total bike rentals.

The correlation was:

-0.2345

This indicates a **weak negative relationship**.

Higher windspeed was associated with slightly lower bike rental counts in the dataset.

However, windspeed alone does not explain the complete variation in rental demand.

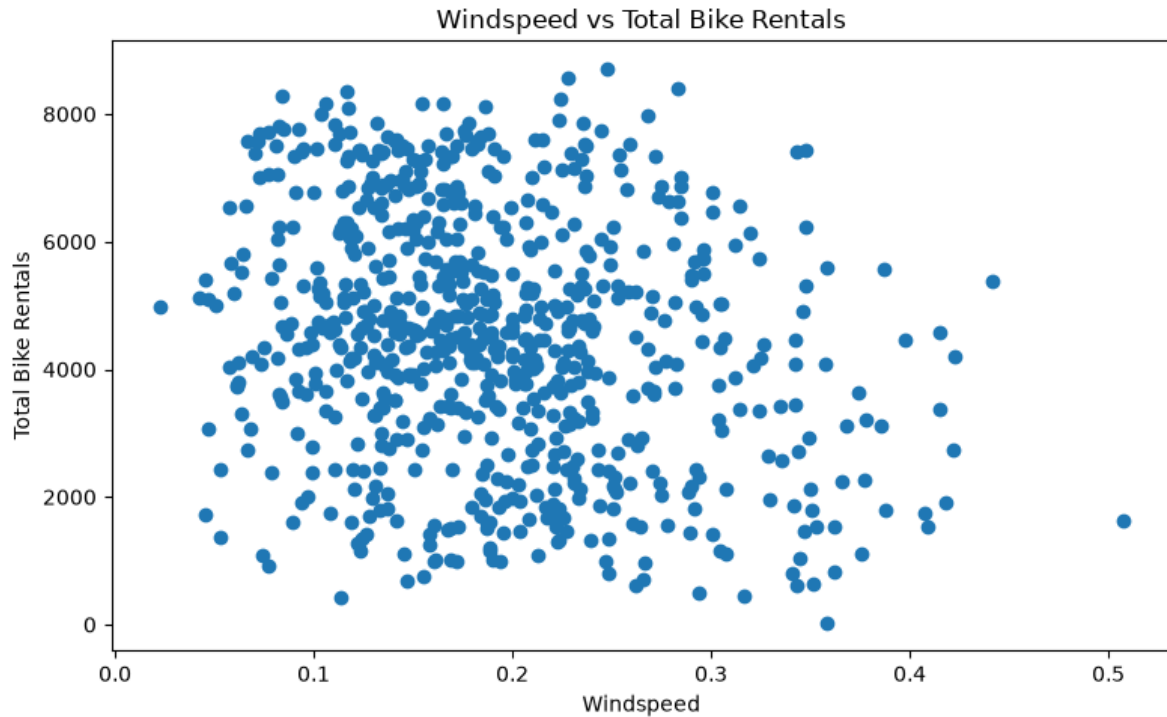


Figure 8.3: Windspeed vs Total Bike Rentals

8.4 Weather Situation vs Total Bike Rentals

A boxplot was used to compare total bike rentals under different weather situations.

The weather situation variable contains codes from 1 to 4. However, the daily dataset used for the main analysis contains observations for weather situations **1, 2, and 3**, while there are no daily observations for situation 4.

The average rental demand was different across the recorded weather situations.

This indicates an association between weather conditions and bike rental demand.

However, the analysis does not prove that weather conditions alone cause changes in rental demand.

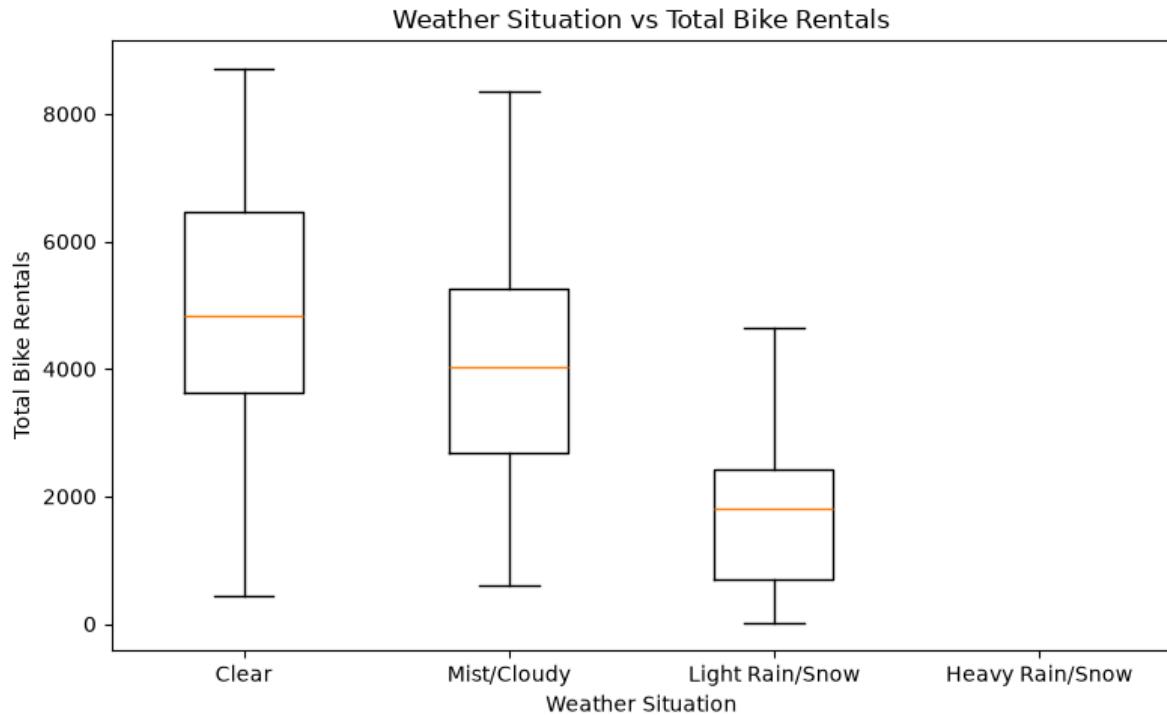


Figure 8.4: Weather Situation vs Total Bike Rentals

8.5 Working Day vs Total Bike Rentals

A boxplot was used to compare bike rentals on working days and non-working days.

The average number of rentals was:

- **Working day: 4584.82**
- **Non-working day: 4330.17**

The average was somewhat higher on working days in this dataset.

However, working-day status alone does not explain the complete variation in bike rental demand.

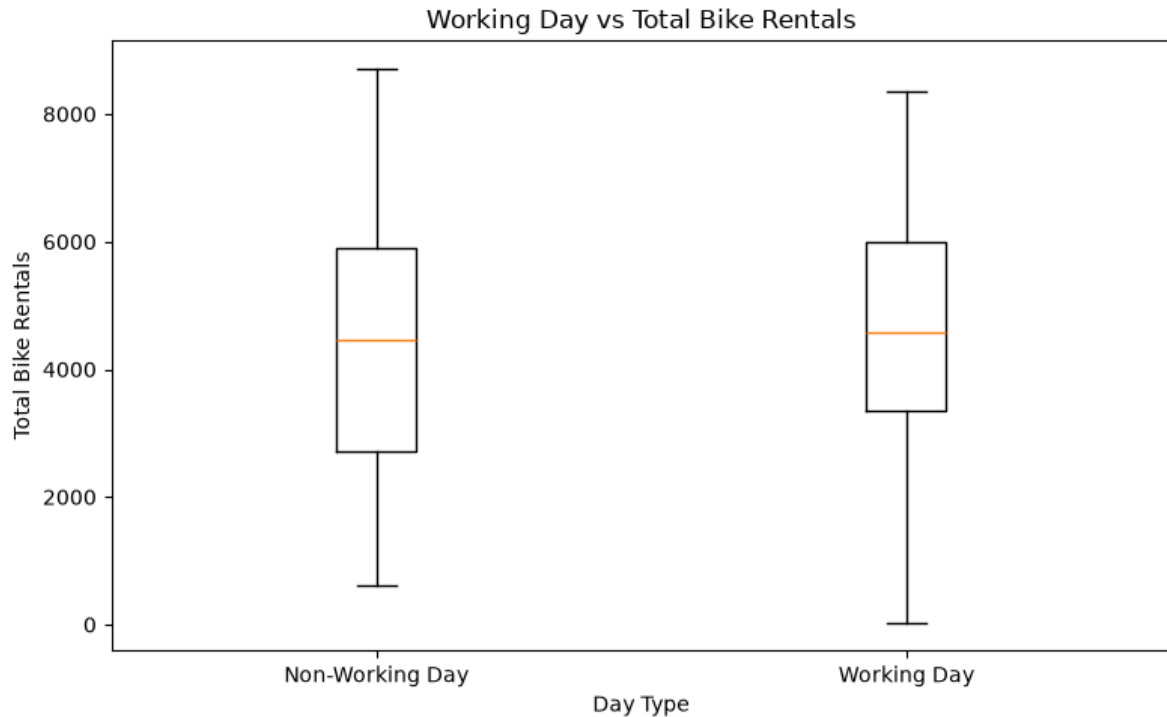


Figure 8.5: Working Day vs Total Bike Rentals

9. Grouping and Aggregation

Grouping and aggregation were used to examine how bike rental demand varied across seasons, months, years, and weather situations.

The Pandas `groupby()` function and mean aggregation were used.

9.1 Rentals by Season

Season	Average Rentals
--------	-----------------

Season 1	2604.13
----------	---------

Season 2	4992.33
----------	---------

Season 3	5644.30
----------	---------

Season 4	4728.16
----------	---------

Season 3 had the highest average number of bike rentals, while Season 1 had the lowest.

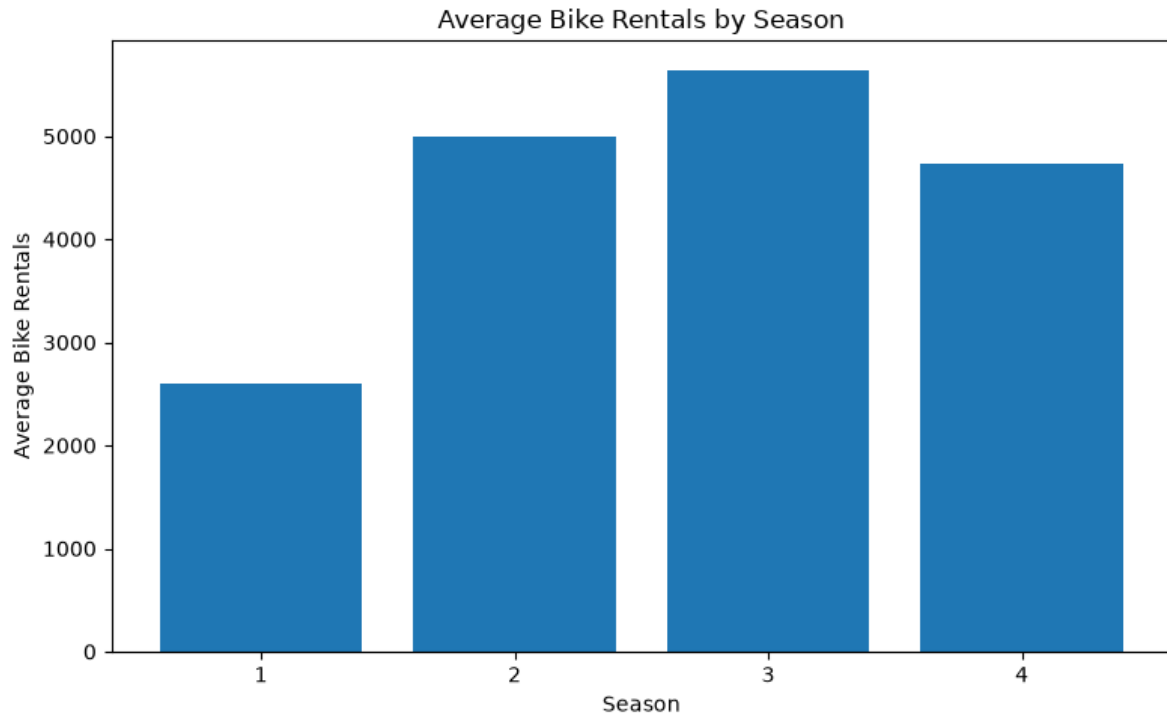


Figure 9.1: Average Bike Rentals by Season

9.2 Rentals by Month

The highest average rental count was recorded in **June**, with **5772.37 rentals**.

The lowest average was recorded in **January**, with **2176.34 rentals**.

This shows considerable variation in rental demand across months.

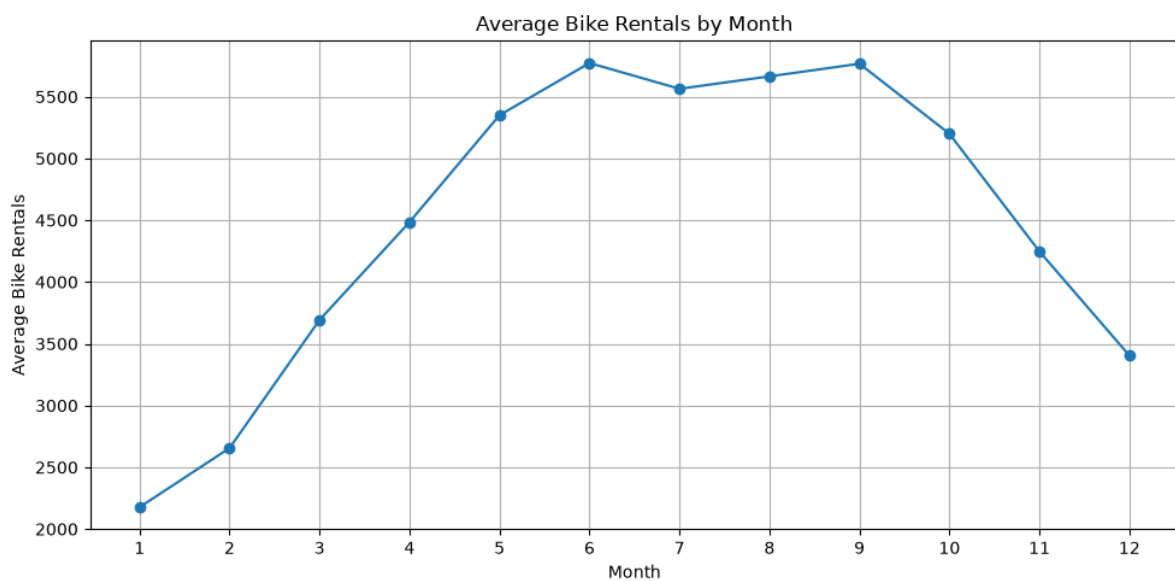


Figure 9.2: Average Bike Rentals by Month

9.3 Rentals by Year

The yr variable uses:

- 0 = 2011
- 1 = 2012

Year Average Rentals

2011 3405.76

2012 5599.93

The average number of daily rentals was higher in 2012 than in 2011.

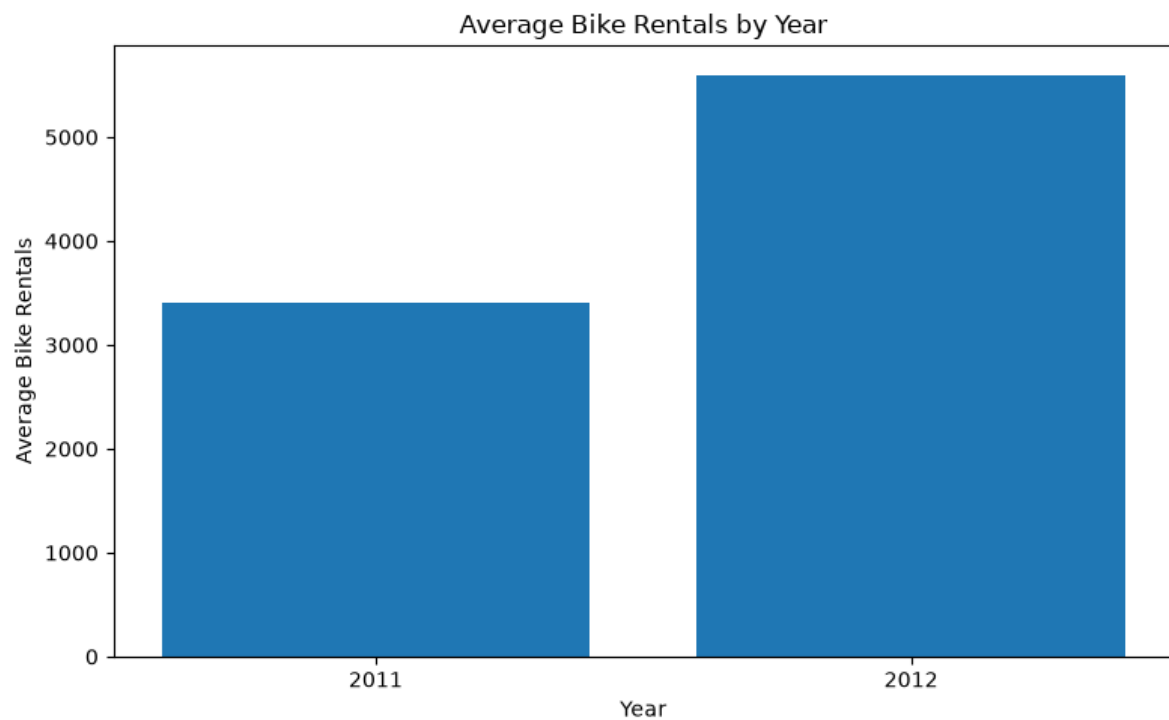


Figure 9.3: Average Bike Rentals by Year

9.4 Rentals by Weather Situation

Weather Situation	Average Rentals
-------------------	-----------------

1 – Clear/Partly Cloudy	4876.79
-------------------------	---------

2 – Mist/Cloudy	4035.86
-----------------	---------

3 – Light Rain/Snow	1803.29
---------------------	---------

The highest average rental demand was observed during Weather Situation 1, while the lowest was observed during Weather Situation 3.

There were no daily observations for Weather Situation 4, so no average was calculated for this category.

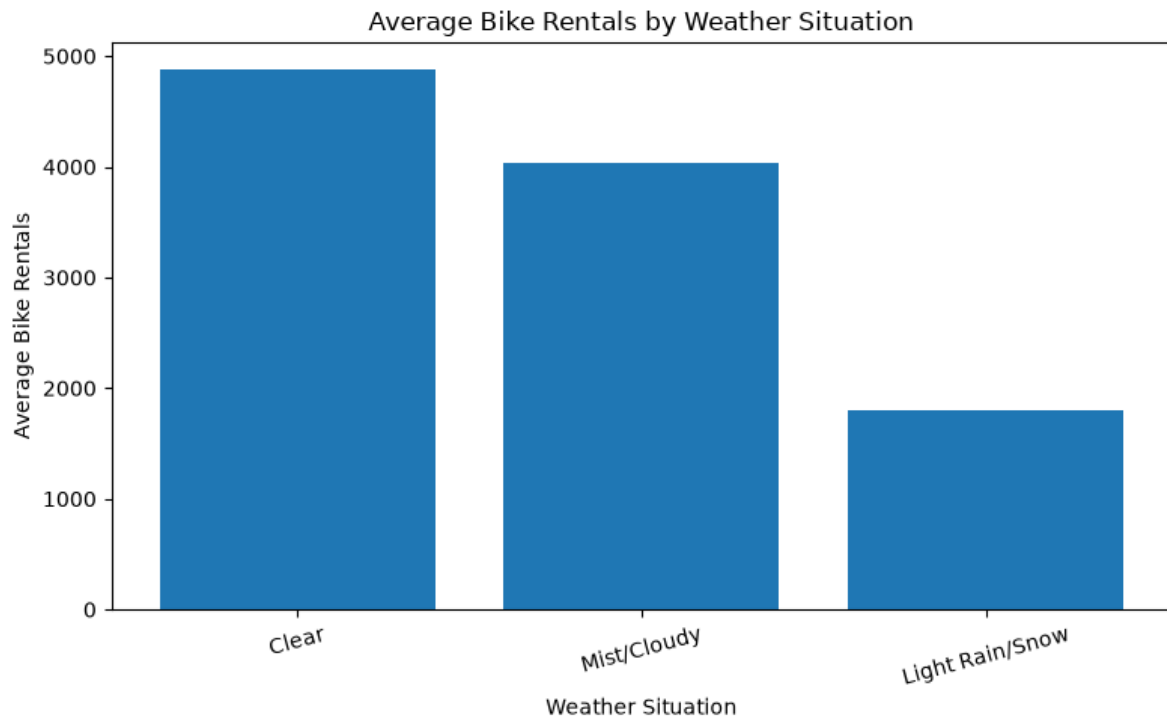


Figure 9.4: Average Bike Rentals by Weather Situation

9.5 Overall Observation

The grouping and aggregation analysis shows that bike rental demand varies across season, month, year, and weather situation.

10. Correlation Analysis

Correlation analysis was performed to study the strength and direction of relationships between important numerical variables.

The Pearson correlation coefficient was used.

A value close to **+1** indicates a strong positive relationship, a value close to **-1** indicates a strong negative relationship, and a value close to **0** indicates a weak linear relationship.

10.1 Important Correlations with Total Bike Rentals

Variable	Correlation with cnt Interpretation	
Temperature (temp)	0.6275	Moderate positive relationship

Variable	Correlation with cnt	Interpretation
Feeling Temperature (atemp)	0.6311	Moderate positive relationship
Humidity (hum)	-0.1007	Weak negative relationship
Windspeed	-0.2345	Weak negative relationship
Casual users	0.6728	Positive relationship
Registered users	0.9455	Very strong positive relationship

Temperature and feeling temperature showed moderate positive relationships with total bike rentals.

Humidity and windspeed showed weak negative relationships with total bike rentals.

The correlation between registered and cnt was very high at **0.9455**.

This is expected because:

cnt = casual + registered

Therefore, this high correlation should not be interpreted as evidence that registered users independently cause an increase in total rentals.

10.2 Temperature and Feeling Temperature

The correlation between temp and atemp was:

0.9917

This indicates a very strong positive relationship.

This is expected because atemp represents feeling temperature and is closely related to temperature.

10.3 Other Relationships

The correlation between casual and registered users was **0.3953**, indicating a moderate positive relationship.

The correlation between windspeed and registered users was **-0.2174**, indicating a weak negative relationship.

The correlation between humidity and registered users was **-0.0911**, indicating a very weak negative relationship.

10.4 Overall Observation

The correlation analysis shows that temperature-related variables have positive relationships with bike rental demand, while humidity and windspeed show weaker negative relationships.

Correlation describes association between variables and does not prove that one variable causes changes in another.

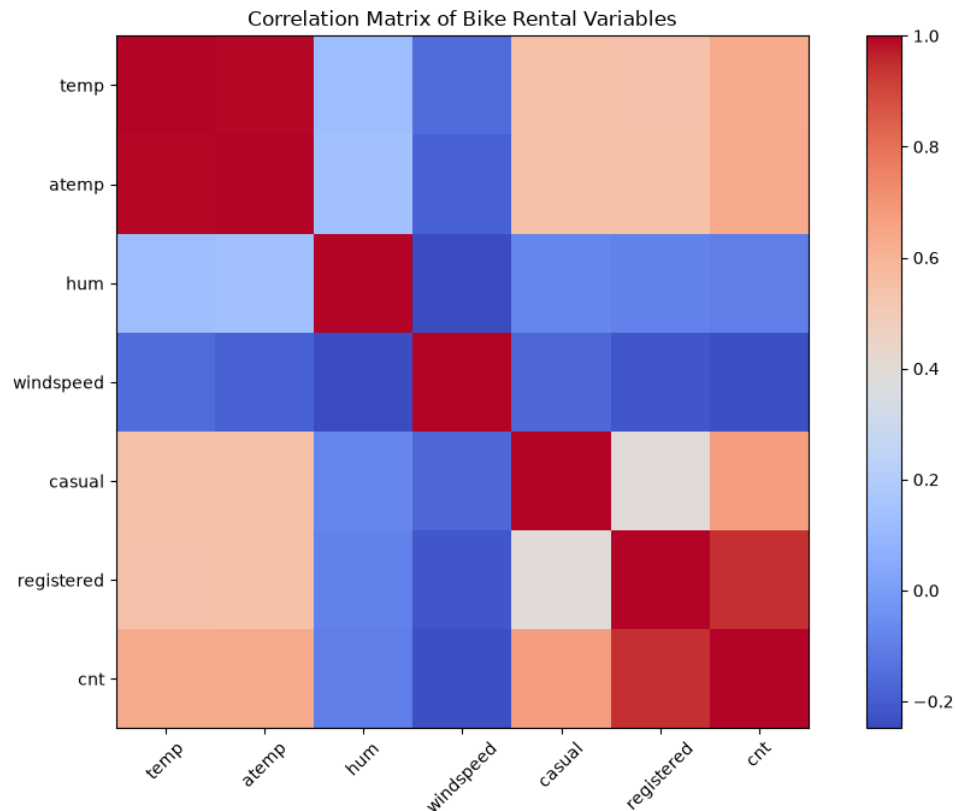


Figure 10.1: Correlation Matrix Heatmap of Bike Rental Variables

11. Visualization

Data visualization was used to present the analysis results in graphical form.

The following visualizations were created:

1. Average Bike Rentals by Month
2. Average Bike Rentals by Season
3. Average Bike Rentals by Year
4. Average Bike Rentals by Weather Situation
5. Temperature vs Total Bike Rentals
6. Correlation Matrix Heatmap

11.1 Average Bike Rentals by Month

A line chart was used to show average bike rentals for each month.

January had the lowest average rental count at **2176.34**, while June had the highest at **5772.37**.

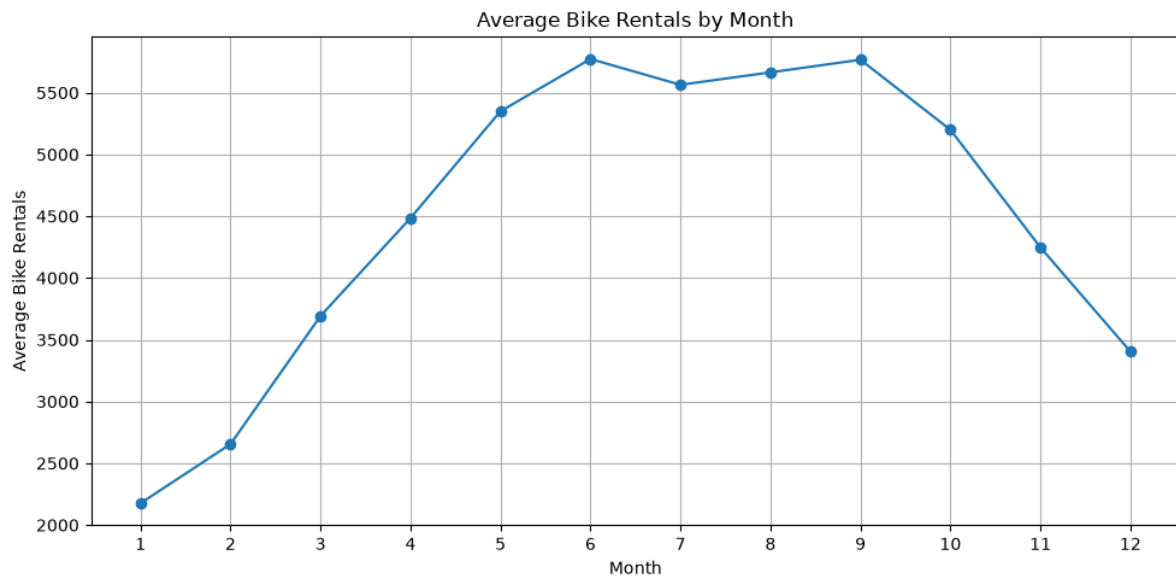


Figure 11.1: Average Bike Rentals by Month

11.2 Average Bike Rentals by Season

A bar chart was used to compare average bike rentals across seasons.

- Season 1: 2604.13
- Season 2: 4992.33
- Season 3: 5644.30
- Season 4: 4728.16

Season 3 had the highest average rental count.

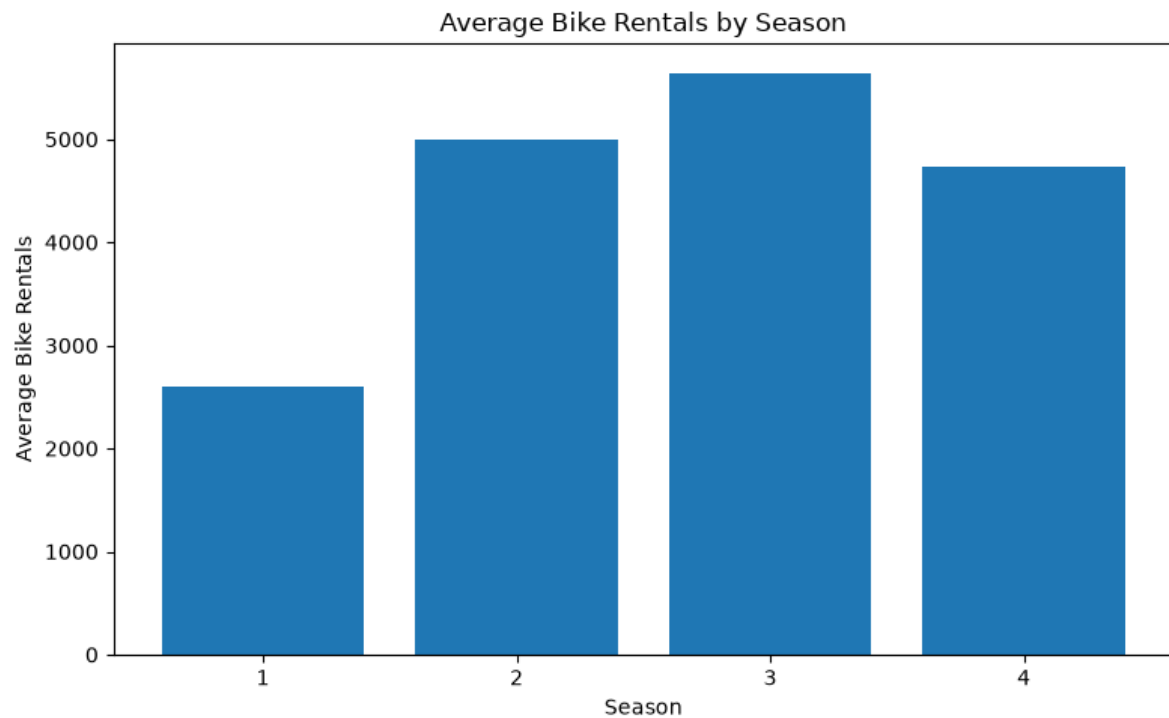


Figure 11.2: Average Bike Rentals by Season

11.3 Average Bike Rentals by Year

A bar chart was used to compare average rentals between 2011 and 2012.

- 2011: 3405.76
- 2012: 5599.93

The average daily rental count was higher in 2012.

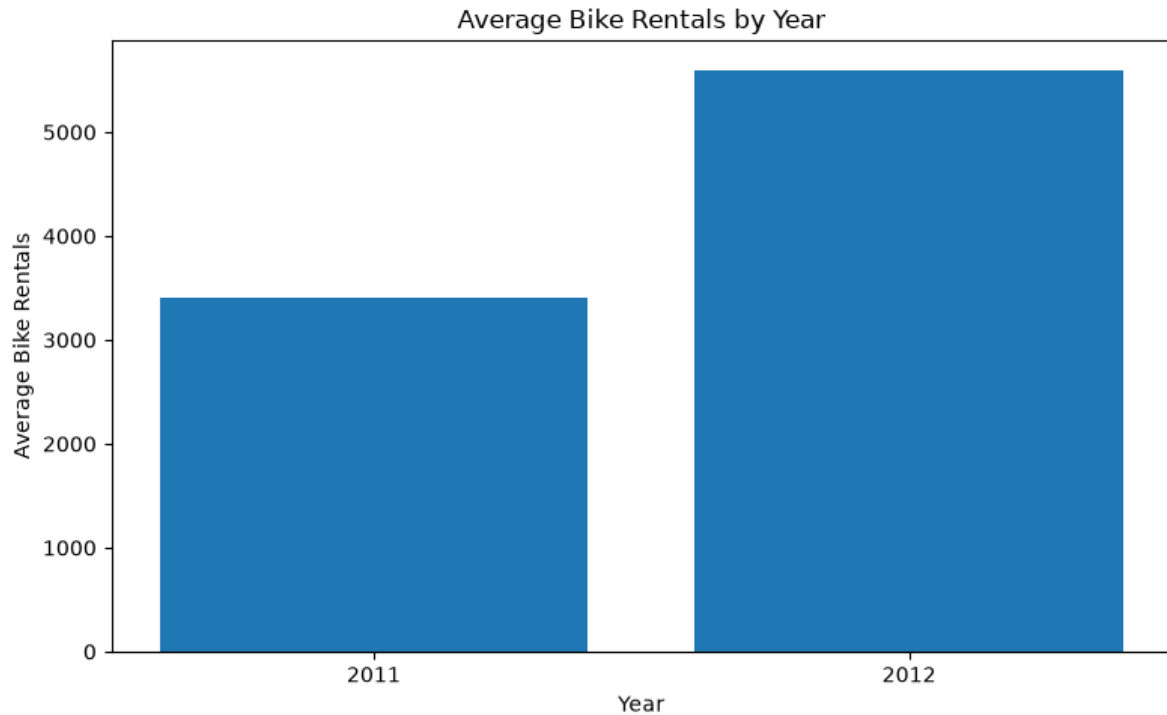


Figure 11.3: Average Bike Rentals by Year

11.4 Average Bike Rentals by Weather Situation

A bar chart was used to compare average rentals under different weather situations.

- Weather Situation 1: 4876.79
- Weather Situation 2: 4035.86
- Weather Situation 3: 1803.29

The lowest average rental demand among the recorded daily weather situations was observed under Weather Situation 3.

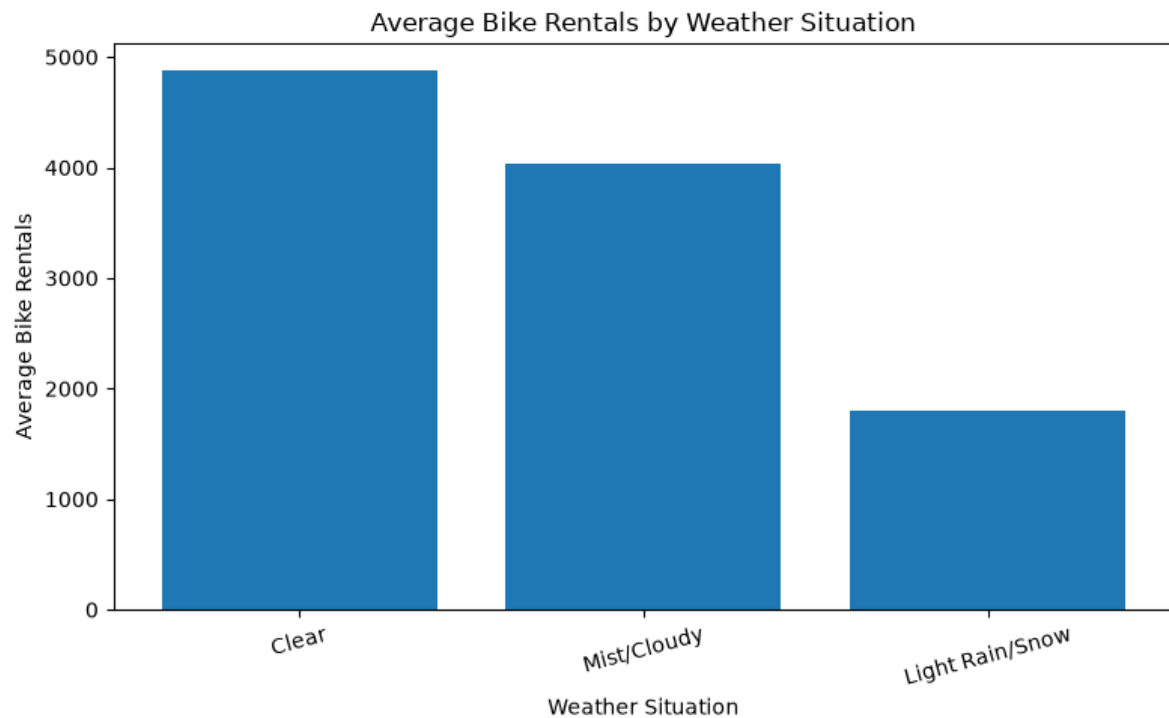


Figure 11.4: Average Bike Rentals by Weather Situation

11.5 Temperature vs Total Bike Rentals

A scatter plot was used to visualize the relationship between temperature and total bike rentals.

The plot shows a general positive relationship.

The correlation was **0.6275**, indicating a moderate positive relationship.

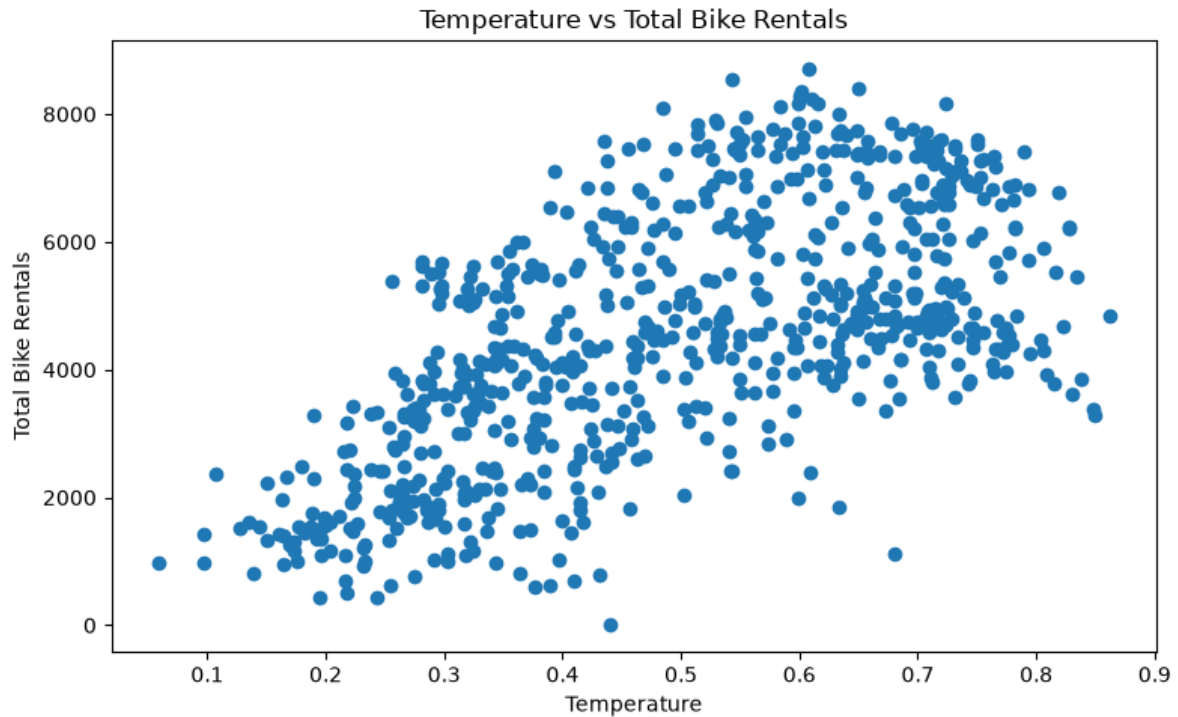


Figure 11.5: Temperature vs Total Bike Rentals

11.6 Correlation Matrix Heatmap

A correlation matrix heatmap was created to visualize relationships between important numerical variables.

The heatmap shows:

- temp and atemp → **0.9917**
- registered and cnt → **0.9455**
- temp and cnt → **0.6275**
- hum and cnt → **-0.1007**
- windspeed and cnt → **-0.2345**

The high correlation between registered and cnt is expected because registered users are one component of total rentals

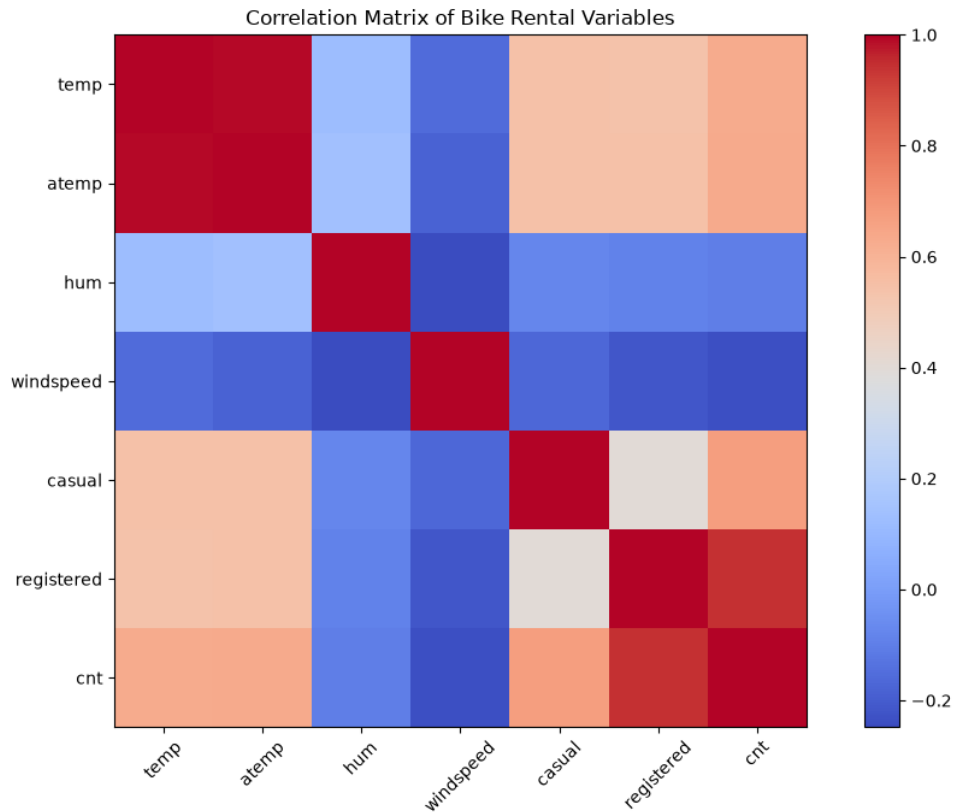


Figure 11.6: Correlation Matrix Heatmap

11.7 Overall Visualization Observation

The visualizations make it easier to identify changes in bike rental demand across months, seasons, years, weather conditions, and temperature.

The graphical results support the findings obtained from descriptive statistics, bivariate analysis, grouping, and correlation analysis.

12. Key Findings

The main findings from the Exploratory Data Analysis are:

1. Dataset Quality:

The day dataset contains **731 records and 16 variables**, while the hour dataset contains **17,379 records and 17 variables**. No missing values or duplicate rows were found.

2. Data Consistency:

The total rental count was checked using $\text{casual} + \text{registered} = \text{cnt}$, and no incorrect total rows were found in the hourly dataset.

3. **Overall Bike Rental Demand:**

The average daily bike rental count was **4504.35**, with a minimum of **22** and a maximum of **8714**.

4. **User Types:**

Registered users accounted for a larger average number of rentals than casual users. The average registered-user rentals were **3656.17**, compared with **848.18** casual-user rentals.

5. **Temperature:**

Temperature had a correlation of **0.6275** with total bike rentals, indicating a moderate positive relationship.

6. **Humidity:**

Humidity had a correlation of **-0.1007**, indicating a weak negative relationship with total rentals.

7. **Windspeed:**

Windspeed had a correlation of **-0.2345**, indicating a weak negative relationship.

8. **Seasonal Pattern:**

Season 3 had the highest average rentals at **5644.30**, while Season 1 had the lowest at **2604.13**.

9. **Monthly Pattern:**

June had the highest average rental demand at **5772.37**, while January had the lowest at **2176.34**.

10. **Yearly Pattern:**

Average daily rentals were **3405.76 in 2011** and **5599.93 in 2012**.

11. **Weather Pattern:**

Weather Situation 1 had the highest average rental demand at **4876.79**, while Weather Situation 3 had the lowest at **1803.29** among the weather situations represented in the daily dataset.

12. **Working Day Pattern:**

Average rentals were **4584.82 on working days** and **4330.17 on non-working days**.

13. **Correlation Analysis:**

The strongest correlation among the selected variables was between temp and atemp, at **0.9917**. The correlation between registered and cnt was **0.9455**, which is expected because registered users are included in total rentals.

14. **Overall Observation:**

Bike rental demand varied across temperature, season, month, year, weather

situation, and working-day status. These variables were associated with changes in rental demand, but the analysis does not establish that any single variable caused those changes.

13. Conclusion

The Exploratory Data Analysis of the Bike Sharing Dataset was performed to understand patterns and relationships associated with bike rental demand.

The analysis showed that bike rental demand varied across months, seasons, years, and weather conditions.

Temperature showed a moderate positive relationship with total bike rentals, while humidity and windspeed showed weaker negative relationships.

The analysis also showed that average daily bike rentals were higher in 2012 than in 2011. Rental demand also varied considerably across different months and seasons.

The dataset was checked for missing values, duplicate records, data types, invalid categorical values, and consistency of total rental counts. No missing values or duplicate records were found, and the total rental count was consistent with the sum of casual and registered users.

Different EDA techniques were used, including:

- Descriptive statistics
- Univariate analysis
- Bivariate analysis
- Grouping and aggregation
- Correlation analysis
- Data visualization

Overall, this project provided practical experience in using Python and Pandas to analyze real-world data and using Matplotlib to visualize patterns and relationships.

The analysis demonstrates how Exploratory Data Analysis can be used to understand changes in bike rental demand based on time, weather, and other recorded factors.

14. References

1. **UCI Machine Learning Repository — Bike Sharing Dataset**
Used as the main source of the bike rental dataset.

UCI Bike Sharing Dataset:

<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

2. Python Documentation

Used as a reference for Python programming concepts.

<https://docs.python.org/3/>

3. Pandas Documentation

Used as a reference for data loading, DataFrame operations, cleaning, grouping, aggregation, and statistical analysis.

<https://pandas.pydata.org/docs/>

4. Matplotlib Documentation

Used as a reference for creating graphs and visualizations.

<https://matplotlib.org/stable/>

15. Code / GitHub

The complete Exploratory Data Analysis was performed using Python.

The Python code was used for:

- Loading the datasets
- Checking data quality
- Converting date values
- Calculating descriptive statistics
- Performing univariate analysis
- Performing bivariate analysis
- Grouping and aggregation
- Correlation analysis
- Creating visualizations

The main Python libraries used were:

- **Pandas** — data loading, cleaning, grouping, aggregation, and analysis
- **Matplotlib** — data visualization

The main Python file is:

EDA.py

The datasets used are:

- **day.csv**
- **hour.csv**

Project Structure

Bike Rental EDA Project/

```
|
|
|— EDA.py
|— day.csv
|— hour.csv
|
|— Report/
    |— Bike Rental EDA Report.pdf
```

GitHub Repository

The project can be uploaded to GitHub for storing and sharing the source code.

GitHub Repository: To be added after uploading the project.

After uploading the project, the GitHub repository link should be added here.

End of Report